

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&E) (Credit System) Degree Examinations**

January 2023

20EEE211 – PROGRAMMABLE LOGIC CONTROLLERS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.

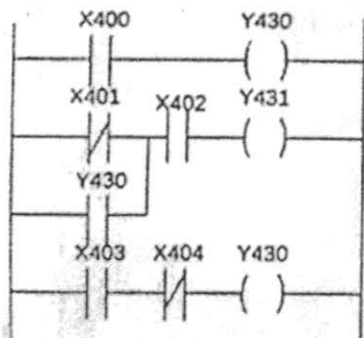
2) Use **Mitsubishi** address notation only.

Unit – I**Marks BT* CO* PO***

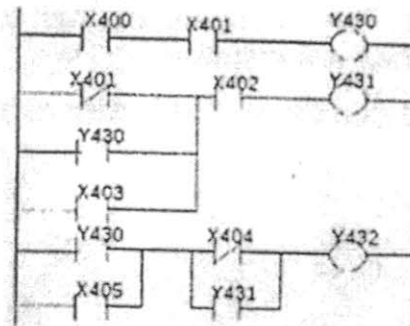
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|--|---|-----|---|---|
| 1. a) Explain the characteristics of the relay, transistor and triac types of PLC output channels with appropriate diagrams. | 8 | L*2 | 1 | 2 |
| b) Describe the parameters to be specified for successful serial communication. | 6 | L1 | 1 | 1 |
| c) Compare different network topologies that are Used for PLC system implementation. | 6 | L1 | 1 | 1 |
| 2. a) With the help of neat diagram, explain the function of each block of internal architecture of PLC. | 8 | L2 | 1 | 1 |
| b) Discuss on proximity switches. Explain different types of proximity switches. | 6 | L1 | 1 | 1 |
| c) Describe sourcing and sinking concepts with reference to I/O units of PLC using appropriate figures. | 6 | L2 | 1 | 2 |

Unit – II

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|--|---|----|---|---|
| 3. a) Draw the functional blocks with timing diagrams for (i) Three input AND gate (ii) Three input NAND gate (iii) XOR gate (iv) XNOR gate. | 8 | L2 | 2 | 2 |
| b) Design a motor controller that has a forward and a reverse button with one stop push button. The motor forward and reverse outputs will only be on when one of the buttons is pushed. When both buttons are pushed, the motor will not work. Implement using LAD. | 6 | L3 | 3 | 3 |
| c) Write IL for the given Ladder diagram in 3c (i) and 3c (ii). | | | | |



3c (i)



3c (ii)

6 L2 2 2

P.T.O.

- | | | | | | |
|-------|---|---|----|---|---|
| 4. a) | Explain with an example how a stop switch and emergency stop switch can be located to ensure a safe system. | 8 | L2 | 3 | 2 |
| b) | Draw the ladder diagram and FBD for the case of a valve which is to be operated to lift a load when the pump is running and either the lift switch is operated or a switch operated indicating that the load has not already been lifted and it is at the bottom of its lift channel. | 6 | L3 | 3 | 3 |
| c) | Briefly explain various structured text operators used in PLC system. | 6 | L2 | 2 | 2 |
| 5. a) | Explain sequential function chart. Explain the switching sequence of the general traffic lights system using SFCs. | 8 | L2 | 2 | 2 |
| b) | Define the term latching circuit. Explain its significance with an example. | 6 | L2 | 3 | 2 |
| c) | Explain the principle of a master control relay with an example. Also write the instruction list for the same. | 6 | L2 | 3 | 2 |

Unit – III

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|-------|--|----|----|---|---|
| 6. a) | Explain different data handling functions available in PLC with example ladder rungs for each. | 10 | L2 | 5 | 2 |
| b) | Explain ladder diagram developed for a 4-bit shift register. Indicate timing diagram of different inputs and outputs. | 10 | L2 | 5 | 2 |
| 7. a) | With a neat block diagram and corresponding ladder diagram, explain the central heating system. | 10 | L2 | 5 | 2 |
| b) | Explain the significance of counters in the industrial process control. Also explain what is pulse timer with the help of ladder diagram and timing diagram. | 10 | L2 | 4 | 2 |
| 8. a) | What is closed loop control system? Explain how a PLC can be used in closed loop control system, with the help of a neat block diagram. | 10 | L2 | 5 | 2 |
| b) | Develop a ladder diagram for the packing system of bottles. A sensor is deployed to check the correctness of liquid level inside the bottle, such that a reject mechanism is activated to deflect the bottles with incorrect liquid levels. Use shift register to perform the required task. | 10 | L4 | 5 | 4 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
